

Po-Shao Chen

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SUMMARY

PhD candidate in ECE with 4 silicon tape-outs (Intel 16, TSMC 28nm, TSMC 40nm $\times 2$) plus one in progress (TSMC 16nm), spanning robotics, GNN, image processing, and communications. Experienced in RTL design, microarchitecture, synthesis, and physical design. Seeking full-time RTL/ASIC design roles starting 2027.

EDUCATION

Doctor of Philosophy in Electrical and Computer Engineering Sep. 2022 – May 2027 (Exp.)
University of Michigan *Ann Arbor, Michigan*

Master of Science in Electronics Engineering Sep. 2018 – Feb. 2021
National Taiwan University *Taipei, Taiwan*

- Thesis: An Energy-Efficient Accelerator IC for Dark Channel Prior Based Blind Image Deblurring

Bachelor of Science in Electrical Engineering Sep. 2014 – June 2018
National Taiwan University *Taipei, Taiwan*

SKILLS

Hardware & Software Languages: SystemVerilog, Verilog, Python, C/C++, MATLAB

EDA & Design Tools: Design Compiler, Innovus, Verdi/nWave, Calibre, VCS, Virtuoso, Quartus Prime

Design: microarchitecture, pipelining, timing closure, low-power design, RTL-to-GDS flow

RESEARCH & WORK EXPERIENCE

Graduate Student Research Assistant Aug. 2022 – May 2027 (Exp.)
LEAPS, University of Michigan *Ann Arbor, Michigan, USA*

- **3DGS Path Planning Accelerator (TSMC 16nm Silicon Tape-out, in progress)**
 - * Reformulated the path planning algorithm into a hardware-friendly form to reduce memory access and computational complexity
 - * Designing the architecture and RTL (SystemVerilog) for the path planning accelerator
- **Spatial GNN Accelerator (Intel 16 Silicon Tape-out)**
 - * Designed the architecture and RTL (SystemVerilog) for subgraph clustering and weight-access optimization to maximize aggregation throughput
 - * Achieved a $4.8\times$ throughput improvement and a $3.1\times$ increase in compute density over prior state-of-the-art designs
- **MU-MIMO & LDPC Digital Annealer (TSMC 28nm ASIC Design Flow)**
 - * Achieved near-ML MU-MIMO detection with 7 orders of magnitude lower complexity than ML, and proposed a new LDPC cost function that outperforms prior QUBO formulations
 - * Designed the RTL and synthesized the annealers in TSMC 28nm at 200 MHz, achieving $>100\times$ higher LDPC throughput than a prior digital annealer
- **RXO-LDPC: Oscillator-Based LDPC Soft Decoder (TSMC 28nm Silicon Tape-out)**
 - * Designed the layout of the twin-crossbar feedback network mapping the LDPC H-matrix to six-body spin-clause couplings, with a SKILL script that generates it automatically from any H-matrix

- * Reduced maximum feedback-path delay by 22% and delay variation by 40% using alternating metal layers and length-matched wires, verified through parasitic extraction

Research Assistant

Aug. 2021 – July 2022

Digital Circuits and Systems Lab, National Taiwan University

Taipei, Taiwan

• Hybrid Post-Quantum Crypto SoC (TSMC 40nm Silicon Tape-out)

- * Performed placement and routing for the full design and resolved DRC/LVS violations with Calibre for tape-out

Mandatory Military Service

Apr. 2021 – July 2021

R.O.C. Army

Taiwan

Graduate Student Research Assistant

July 2018 – Mar. 2021

Digital Circuits and Systems Lab, National Taiwan University

Taipei, Taiwan

• Blind Image Deblurring Accelerator IC (TSMC 40nm Silicon Tape-out)

- * Analyzed the mathematical frameworks and microarchitectural bottlenecks of the dark channel prior deblurring algorithm
- * Designed the architecture and RTL (Verilog) and taped out a cell-based ASIC achieving $2562\times$ faster full-HD image deblurring compared to a desktop CPU
- * Optimized architecture to achieve $4\times$ higher area efficiency and $7.5\times$ higher energy efficiency than prior designs

JOURNAL PAPERS

L.-H. Lin, Z.-S. Fu, P.-S. Chen, B.-Y. Yang, and C.-H. Yang, “A 4.8mW Hybrid Crypto SoC with Post-Quantum Cryptography and Lightweight Cryptography for Secure Neural Interfacing,” *IEEE Journal of Solid-State Circuits (JSSC)*, May 2026.

E. Dikopoulos, L. Wormald, Y.-T. Hsu, W. Tang, P.-S. Chen, Z. Zhang, M. P. Flynn, “RXO-LDPC: A Physics-Inspired Relaxation Oscillator-Based Solver Leveraging 6-body Spin Interactions for Soft Decoding of LDPC Codes,” *IEEE Journal of Solid-State Circuits (JSSC)*, vol. 60, no. 7, pp. 2420-2433, July 2025.

P.-S. Chen, Y.-L. Chen, Y.-C. Lee, Z.-S. Fu, C.-H. Yang, “A 28.8mW Accelerator IC for Dark Channel Prior Based Blind Image Deblurring,” *IEEE Journal of Solid-State Circuits (JSSC)*, vol. 59, no. 6, pp. 1899-1911, June 2024.

CONFERENCE PAPERS

P.-S. Chen, M.-C. Weng, W. Tang, Z. Zhang, “SGX: A 16nm 1.92TFLOPS Spatial GNN Accelerator Using Subgraph-Parallel Processing,” *IEEE European Solid-State Electronics Research Conference (ESSERC)*, 2026, accepted.

P.-S. Chen, W. Tang, Z. Zhang, “Practical MU-MIMO Detection and LDPC Decoding through Digital Annealing,” *IEEE Design, Automation and Test in Europe Conference (DATE)*, Mar. 2025.

E. Dikopoulos, L. Wormald, Y.-T. Hsu, W. Tang, P.-S. Chen, Z. Zhang, M. P. Flynn, “A Relaxation Oscillator-Based Probabilistic Combinatorial Optimization Engine for Soft Decoding of LDPC Codes,” *IEEE European Solid-State Electronics Research Conference (ESSERC)*, Sep. 2024.

L.-H. Lin, Z.-S. Fu, P.-S. Chen, B.-Y. Yang, and C.-H. Yang, “A 4.8mW, 800Mbps Hybrid Crypto SoC for Post-Quantum Secure Neural Interfacing,” *Int. Symposium on VLSI Circuits (VLSI Circuits)*, June 2023.

P.-S. Chen, Y.-L. Chen, Y.-C. Lee, Z.-S. Fu, and C.-H. Yang, “A 28.8mW Accelerator IC for Dark Channel Prior Based Blind Image Deblurring,” *IEEE Asian Solid-State Circuits Conference (A-SSCC)*, Nov. 2021.